CHAPTER 23

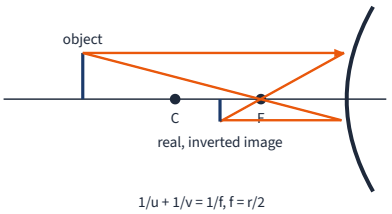
GEOMETRICAL OPTICS

Key idea: mirrors and lenses form images by reflection and refraction; use the real-is-positive convention.

1 SIGN CONVENTION

- Real object, real image: **positive**
- Virtual image: v **negative**
- Concave mirror, convex lens: $f > 0$
- Convex mirror, concave lens: $f < 0$
- $m = -\dfrac{v}{u}$: negative = inverted, positive = upright

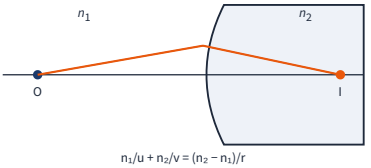
2 SPHERICAL MIRRORS



1/u + 1/v = 1/f, f = r/2

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f} \qquad f=\frac{R}{2}$$

3 REFRACTION AT A CURVED SURFACE



$n_1/u + n_2/v = (n_2 - n_1)/r$

$$\frac{n_1}{u}+\frac{n_2}{v}=\frac{n_2-n_1}{R}$$

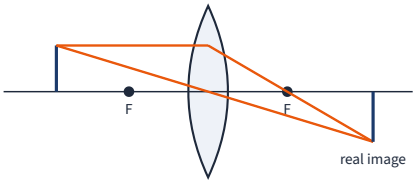
- $R > 0$ if the centre is on the **outgoing** side
- Plane surface: apparent depth $= \dfrac{\text{real depth}}{n}$

4 IMAGE NATURE

Object (converging)	Image
$u > 2f$	real, inverted, diminished
$u = 2f$	real, inverted, same size
$f < u < 2f$	real, inverted, magnified
$u < f$	virtual, upright, magnified

Convex mirror / concave lens: always virtual, upright, diminished.

5 THIN LENSES



1/u + 1/v = 1/f

$$\frac{1}{f}=(n-1)\left(\frac{1}{R_1}+\frac{1}{R_2}\right)$$

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f} \qquad P=\frac{1}{f}$$

Biconvex: both R positive. Power in dioptre (D), f in metres.

6 USEFUL RESULTS

- Lenses in contact: $\dfrac{1}{f}=\dfrac{1}{f_1}+\dfrac{1}{f_2}$, $P = P_1 + P_2$
- Same-size real image: $u = v = 2f$
- Minimum object–image distance: $4f$
- Mirror: f does not depend on the medium; lens: f does

7 RAY-DIAGRAM RULES

1. Ray parallel to the axis → through (or from) F.
2. Ray through the optical centre (lens) or C (mirror) → undeviated / back on itself.
3. Ray through F → emerges parallel to the axis.

8 COMMON SECTION A TRAPS

- Dropping the minus sign of a virtual image or a diverging lens.
- Using $f = R$ instead of $R/2$ for a mirror.
- Wrong sign of R in the lensmaker's or surface equation.
- Mixing cm and m when finding power in D.

EXAM TIPS

- ✓State the sign convention at the start of the answer.
- ✓Draw a quick ray diagram to check the answer is sensible.
- ✓Use $m = -v/u$ and describe the image in three words.
- ✓Find each image in turn for two-lens systems.